

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: GLUE | Context: Luxembourgish Public Administration Civil Servants — GLUE Assessment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

GLUE's input content is entirely English, drawn from American/British movie reviews [\[Q26\]](#), US/UK news [\[Q29, Q48\]](#), Wikipedia [\[Q41\]](#), Quora [\[Q31\]](#), linguistics journals [\[Q21\]](#), and English-language fiction and transcribed speech [\[Q38\]](#). The dataset analysis confirms across 409 sampled examples that 'every single example reviewed across all 12 configs is in English' with US-political (Clinton, Bush, Gingrich, Ryan, Tillerson — [DATASET-D10, D13, D14, D15, D22, D34](#)), India-specific [\[DATASET-D16, D18\]](#), and US-pop-culture [\[DATASET-D3\]](#) cultural references dominating. The deployment input is trilingual Luxembourgish/French/German code-switched citizen correspondence with non-standardized Luxembourgish orthography [\[WEB-12, WEB-11\]](#), from a population that is 47.3% non-Luxembourgish nationals [\[WEB-4\]](#) with 47% of the workforce being cross-border workers from France, Belgium, and Germany [\[WEB-6, WEB-7\]](#). The cultural and linguistic distance is total. IC is marked HIGH priority.

ID	Evidence
Q21	"The Corpus of Linguistic Acceptability (Warstadt et al., 2018) consists of English acceptability judgments drawn from books and journal articles on linguistic theory." (p.3)
Q26	"The Stanford Sentiment Treebank (Socher et al., 2013) consists of sentences from movie reviews and human annotations of their sentiment." (p.3)
Q29	"The Microsoft Research Paraphrase Corpus (Dolan & Brockett, 2005) is a corpus of sentence pairs automatically extracted from online news sources, with human annotations for whether the sentences in the pair are semantically equivalent." (p.3)
Q31	"The Quora Question Pairs dataset is a collection of question pairs from the community question-answering website Quora." (p.3)
Q38	"The premise sentences are gathered from ten different sources, including transcribed speech, fiction, and government reports." (p.4)
Q41	"The Stanford Question Answering Dataset (Rajpurkar et al. 2016) is a question-answering dataset consisting of question-paragraph pairs, where one of the sentences in the paragraph (drawn from Wikipedia) contains the answer to the corresponding question (written by an annotator)." (p.4)
Q48	"Examples are constructed based on news and Wikipedia text." (p.4)
D3	the weird thing about the santa clause 2 , purportedly a children 's movie , is that there is nothing in it to engage children emotionally . — Full sentence from American movie review; US pop-culture reference
D10	I did not mention Monica in my lecture, but the first question I was asked was how President Clinton could do his job with all the distractions caused by the Monica Lewinsky affair. — US political context (Clinton/Lewinsky); culturally US-specific
D13	Platinum prices soared to 23-year highs earlier this year after President Bush (news - web sites) proposed investing \$ 1.2 billion for research on fuel cell-powered vehicles. — US political and financial news paraphrase
D14	Mr. Bush had sought to store his papers in his father's presidential library, where they would have stayed secret for a half-century. — US presidential politics; US-specific cultural/political context
D15	Who succeeded Newt Gingrich as Speaker?" / "In 1998, with Speaker Newt Gingrich announcing his resignation... — US political history QA; requires US-specific knowledge
D16	What is black money and how can it effect the economy of a country?" / "What is the effect of black money on India's macro economy? — Quora question pair; Indian economic/policy context
D18	Are the notes of Rs. 2000 really embedded with a GPS chip?" / "How does the embedded NGC technology of the Rs. 2000 note works? — India-specific currency question; non-EU political context

D22	House Speaker Paul Ryan was facing problems uniquely from fellow Republicans dissatisfied with his leadership." / "House Speaker Paul Ryan was facing problems uniquely from fellow Republicans supportive of his leadership. — US political content in diagnostic set
D34	The announcement of Tillerson's departure sent shock waves across the globe." / "People across the globe were prepared for Tillerson's departure. — US political news (Rex Tillerson/Trump administration) in diagnostic set
WEB-4	47.3% foreign nationals as of 1 January 2024; over 170 nationalities — confirms input population's linguistic diversity (https://luxembourg.public.lu/en/society-and-culture/population/demographics.html)
WEB-6	~47% of workforce are frontaliers — establishes the frontier-vs-resident label requirement (https://statistiques.public.lu/en/publications/series/regards/2025/regards-01-25.html)
WEB-7	https://www.oecd.org/en/publications/2025/04/oecd-economic-surveys-luxembourg-2025_3eb782b5/full-report/reviving-productivity-growth_4509ab88.html
WEB-11	LuxBorrow corpus with token-level LID and code-switching annotation across LU/DE/FR/EN — confirms code-switching as systematic input characteristic (https://arxiv.org/html/2603.10789)
WEB-12	Neural Text Normalization for Luxembourgish — best ByT5 base achieves ~78.8% accuracy; documents 'large amount of orthographic variation' (https://arxiv.org/html/2412.09383v1)

Strengths

- MNLI's deliberate genre diversity — ten sources including government reports, transcribed speech, and fiction [Q38] — at least demonstrates a methodological awareness that NLU evaluation should span register variation, a principle transferable to a custom Luxembourgish suite even though GLUE itself does not realize it for the target language.
- The diagnostic set draws on naturally-occurring sentences from multiple domains [Q63], offering a genre-diversity template for a custom Luxembourgish administrative diagnostic.

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Q38	"The premise sentences are gathered from ten different sources, including transcribed speech, fiction, and government reports." (p.4)
Q63	"We ensure the data is reasonably diverse by producing examples for a variety of linguistic phenomena and basing our examples on naturally-occurring sentences from several domains (news, Reddit, Wikipedia, academic papers)." (p.5)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires region-specific knowledge including Luxembourgish lexicon and orthographic variation [WEB-12, WEB-15], French/German/Luxembourgish code-switching patterns [WEB-11], frontier-vs-resident terminology, EU institutional vocabulary, communal-vs-État administrative terminology, and continental-European formal correspondence conventions. None of these are present in GLUE [Q21, Q26, Q29, Q31, Q35, Q41].

IC-2: Cultural alignment is absent. The dominant cultural content is American (US politics, US sports, US movies, US-centric Quora threads — DATASET-D10, D13, D14, D17, D20, D22, D34, D42) with some Indian content [DATASET-D16, D18]. No Luxembourgish or even Luxembourg-adjacent (Belgian/French frontier) administrative content appears in the 409-example sample.

IC-3: Yes — Western-specific knowledge that does not transfer is pervasive: US presidential history [DATASET-D10, D14], US congressional politics [DATASET-D15, D22], American film criticism conventions [DATASET-D1, D2, D3, D4, D29, D30], antebellum American fiction dialect [DATASET-D11, D28], American telephone-speech register [DATASET-D33], and English-specific morphosyntactic violations [DATASET-D8, D40].

IC-4: INSUFFICIENT DOCUMENTATION — GLUE does not document recruitment of regional annotators, and no Luxembourgish annotator pool exists in the inherited datasets. The YAML notes that no Luxembourgish administrative-correspondence corpus is publicly available and CTIE access would be required.

IC-5: The content-validity violation is severe: the input distribution shares neither language, script-norm domain, register, nor cultural reference frame with the deployment. Construct-irrelevant variance from US/Indian cultural specificity will dominate any signal a model carries from GLUE.

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Q31	"The Quora Question Pairs dataset is a collection of question pairs from the community question-answering website Quora." (p.3)
Q35	"The Semantic Textual Similarity Benchmark (Cer et al., 2017) is a collection of sentence pairs drawn from news headlines, video and image captions, and natural language inference data. Each pair is human-annotated with a similarity score from 1 to 5; the task is to predict these scores. Follow common practice, we evaluate using Pearson and Spearman correlation coefficients." (p.3)
Q41	"The Stanford Question Answering Dataset (Rajpurkar et al. 2016) is a question-answering dataset consisting of question-paragraph pairs, where one of the sentences in the paragraph (drawn from Wikipedia) contains the answer to the corresponding question (written by an annotator)." (p.4)
D1	sleepwalk through vulgarities in a sequel you can refuse — English movie-review fragment; sentiment label calibrated to film criticism register
D2	at its best moments — Sub-sentential movie-review fragment; context-free without film domain
D3	the weird thing about the santa clause 2 , purportedly a children 's movie , is that there is nothing in it to engage children emotionally . — Full sentence from American movie review; US pop-culture reference
D4	no lika da — Non-standard English fragment (phonetic representation); unclear sentiment signal
D8	They drank the pub. — Idiomatic English usage; grammatical acceptability judgment
D10	I did not mention Monica in my lecture, but the first question I was asked was how President Clinton could do his job with all the distractions caused by the Monica Lewinsky affair. — US political context (Clinton/Lewinsky); culturally US-specific
D11	Yes, Mistuh Reese, suh? — Dialect representation of African American vernacular in fiction
D13	Platinum prices soared to 23-year highs earlier this year after President Bush (news - web sites) proposed investing \$ 1.2 billion for research on fuel cell-powered vehicles. — US political and financial news paraphrase
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D16	What is black money and how can it effect the economy of a country? / "What is the effect of black money on India's macro economy? — Quora question pair; Indian economic/policy context
D17	How do mountain ranges form, and what are some of the major mountain ranges in Oklahoma? / "How do mountain ranges in Oklahoma differ from mountain ranges in Idaho? — US geography; Quora community context
D18	Are the notes of Rs. 2000 really embedded with a GPS chip? / "How does the embedded NGC technology of the Rs. 2000 note works? — India-specific currency question; non-EU political context
D20	In-form Rooney's hot goalscoring streak of seven goals in his last four internationals saw him win the vote to be crowned England's Player of the Year for 2008. — British sports news; no Luxembourg administrative relevance
D22	House Speaker Paul Ryan was facing problems uniquely from fellow Republicans dissatisfied with his leadership." / "House Speaker Paul Ryan was facing problems uniquely from fellow Republicans supportive of his leadership. — US political content in diagnostic set

D28	Yes, Mistuh Reese, suh?" / "The slave spoke to Mr Reece. — Dialect transcription from American fiction (antebellum setting)
D29	the horrors — Two-word fragment; highly context-dependent sentiment label
D30	heroes — Single-word fragment; no sentential context
D33	oh yeah no that's uh that's a that's a real interesting movie and it's got a good historical perspective to it — Transcribed telephone speech; casual American spoken English
D34	The announcement of Tillerson's departure sent shock waves across the globe." / "People across the globe were prepared for Tillerson's departure. — US political news (Rex Tillerson/Trump administration) in diagnostic set
D40	The children eat all chocolate. — English determiner usage; grammaticality judgment
D42	Grisham did not win the popular vote." / "Grisham almost won the popular vote. — US electoral context in diagnostic set
WEB-4	47.3% foreign nationals as of 1 January 2024; over 170 nationalities — confirms input population's linguistic diversity (https://luxembourg.public.lu/en/society-and-culture/population/demographics.html)
WEB-6	~47% of workforce are frontaliers — establishes the frontier-vs-resident label requirement (https://statistiques.public.lu/en/publications/series/regards/2025/regards-01-25.html)
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WEB-12	Neural Text Normalization for Luxembourgish — best ByT5 base achieves ~78.8% accuracy; documents 'large amount of orthographic variation' (https://arxiv.org/html/2412.09383v1)
WEB-15	spellux Python package documents homograph density and orthographic variation as ongoing Luxembourgish NLP challenges (https://github.com/questoph/spellux)

Information Gaps

- No annotator demographic information is published for the inherited GLUE component datasets, so the regional-annotator question cannot be addressed from documentation.

Requires Expert Verification

- Sub-national linguistic variation within Luxembourg (Minett, Ardennes, Moselle, Stad Lëtzebuerg) requires sociolinguistic expert elicitation beyond available web sources.

Remediation

Gap 1: GLUE input content is entirely English (US/UK news, movie reviews, Wikipedia, Quora) with no Luxembourgish, French, or German text and no administrative-domain content.

Recommendation: Build a Luxembourgish administrative-correspondence corpus from MyGuichet.lu data (with appropriate CTIE access and CNPD/GDPR review), supplementing with LuxBorrow [WEB-11] for code-switching patterns and adapting ItzGLUE's content domains. Ensure the corpus reflects the input population's linguistic diversity (47.3% non-Luxembourgish nationals, ~47% frontier workforce).

ID	Evidence
WEB-11	LuxBorrow corpus with token-level LID and code-switching annotation across LU/DE/FR/EN — confirms code-switching as systematic input characteristic (https://arxiv.org/html/2603.10789)